

DS6021: Intro to Predictive Modeling

Lab 0: Data Exploration with Class Data

August 28, 2026

Due: August 30, 2026 11:59pm

Note

This assignment will not be graded. It is intended to get you up and running and used to exploring data in Python, as well as using a standard set of packages and interacting with github. These are things we will do all semester.

Prerequisites

If you haven't already, get yourself setup with git and Python, and clone our classroom repository:

- [DS6021_F26 course repo](#)

In addition, this is a great opportunity to practice turning things in with github, especially for those of you with less experience.

- [Instructions in our course repo](#)
- [Example repository](#)

Dataset

Use the dataset [DS6021-Fall26-Anonymized.csv](#) located in our class github repository. Please note that I have removed everyone's email address, and the answer to the final question, "What is one thing I should know before we start the semester?"

Tasks

1. Data Cleaning

- Carefully inspect every column in the dataset.
- Clean entries so that values are consistent and interpretable. Do this using pandas and Python vs doing it in Excel or Google sheets.
 - Note: Please keep track of the code that you use to do this, so that someone else could start with the raw data, take the same steps, and end up with exactly the same cleaned dataset.
- Ensure that missing values remain only where data was originally missing (no unintended information loss).
- Convert each column into its appropriate data type (e.g., numeric, categorical, text, date).
 - If the default data type selected by Python is different from what you think it should be, please explain why.

2. Univariate & Bivariate Exploration

- (Univariate) Choose 4-5 columns in the dataset and create clear visualizations to explore the distribution of each column individually. Choose appropriate graphs (e.g., histograms, bar charts, box plots) depending on the type of data.
- (Bivariate) Choose at least three pairs of relationships between two variables at a time. One example might be looking at the relationship between Python proficiency and SQL usage. Use visualizations (e.g., scatter plots, grouped bar charts, heatmaps, pair plots) to highlight trends, correlations, or patterns.

3. Reflections

- Is there anything about the dataset that surprised you?
- For some of the survey response questions, there are multiple ways that one could clean the data. For two different examples where you had a choice to make, please discuss the alternatives that you considered and why you made the choice that you ultimately did.
- In the first part of this assignment, I asked you to use Python as opposed to cleaning the data in Excel or Google Sheets. In a business context, what is a potential negative outcome that could occur if you did this in Excel or Google sheets?

Deliverables

- All code used to clean and transform the data, do variable exploration, and generate plots.
- A set of visualizations with captions or notes that describe what you observe.
 - Note: These could be included in a Python notebook or in a separate file.
- A short summary (1–2 paragraphs) of your key findings from the univariate and bivariate analyses, as well as answers to the reflection questions.